4.6 Newton's Method

For thousands of years, one of the main goals of mathematics has been to find solutions to equations. For linear equations

$$ax + b = 0,$$

and for quadratic equations

$$ax^2 + bx + c = 0,$$

we can explicitly solve for a solution. However, for most equations there is no simple formula that gives the solutions.

In this section we study a numerical method called *Newton's method* or the *Newton–Raphson method*, which is a technique to approximate the solutions to an equation $f(x) = 0$.

◆ Procedure for Newton's Method

Suppose that $x = c$ is an unknown zero of $f(x)$ and that $f(x)$ is differentiable in an open interval containing c. To approximate c

Step 1. Make an initial approximation x_0 (guess) close to c.

Step 2. Determine a new approximation using the formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \qquad n = 0,1,2,\cdots$$

Step 3. $x_n \to c$ as $n \to \infty$.

Example 1 Approximate the positive root of the equation

$$f(x) = x^2 - 2 = 0.$$

Example 2 Find the x-coordinate of the point where the curve $y = x^3 - x$ crosses the horizontal line $y = 1$.